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Composition analysis of glasses in the PbBr₂–PbCl₂–PbF₂–PbO–P₂O₅ system

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Abstract

Composition change during the melting process of some glasses in the PbBr₂–PbCl₂–PbF₂–PbO–P₂O₅ system melted at 450–470°C for 15 min was studied. Results show that there was a remarkable difference between the batch composition and the analyzed composition. Large amount of P, Br, Pb, and Cl were lost mainly in the form of PbBr₂, PbCl₂ and P₂O₅. The content of O is influenced by two factors. The incomplete decomposition of NH₄H₂PO₄ or the reaction between P₂O₅ and H₂O in the atmosphere increases the content of O, while the volatilization of P₂O₅ decreases the content of O. © 2001 Elsevier Science B.V. All rights reserved.

1. Introduction

Low melting glasses with good water durability can be used in many fields and there is a growing interest in the optical properties of glasses doped with organic compounds. In order to dope an organic compound into an inorganic glass, the melting temperature of the glass must be lower than the decomposing temperature of the organic compound.

We have prepared some new lead halide glasses in the PbBr₂–PbCl₂–PbF₂–PbO–P₂O₅ system [1] and studied their properties. The lead halide glasses have relatively low glass transition temperature (>146°C) and relatively high density

(5.48 g/cm^{−3}). The glasses have good water durability. The dissolution rate of most of the glasses in deionized water at 25°C is in the order of 10^{−4}–10^{−5} mm/day.

In this paper, the composition change during melting of glass in the PbBr₂–PbCl₂–PbF₂–PbO–P₂O₅ system was studied and the form of the loss of each element is discussed.

2. Experimental approach

Lead bromide (PbBr₂) was prepared in our laboratory in a way described in [2]. The other materials used in the glass preparation were commercially obtained PbO, PbF₂, PbCl₂ and NH₄H₂PO₄. All these chemicals are analytical grade.

The glasses were prepared through a two-step melting process to minimize changes in composition during melting. In the first step, appropriate amounts of NH₄H₂PO₄, PbCl₂, PbO and/or PbF₂

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were put into a porcelain crucible without cover at 500°C for 30 min using an electric furnace. During the process, $\text{NH}_4\text{H}_2\text{PO}_4$ was decomposed according to the following reaction:



Then the crucible was removed from the furnace and cooled in air to near room temperature. In the second step, the required amount of PbBr_2 was added to the crucible. The crucible was put into the furnace and the batches were melted at 450–470°C for 15 min. Glass samples were formed by quenching the melts in a carbon steel mold at room temperature. The sizes of the glass samples are 20 mm in diameter and 5 mm in thickness. The glass samples were annealed in an electric oven at 5–10°C above the glass transition temperature for 4 h and cooled in the oven.

The composition of the glasses was determined by chemical analysis. The amounts of Pb, Cl and Br were determined using wet-chemical analysis; the relative error is $\pm 0.5\%$. The P content was analyzed using a spectrophotometer; the relative error is $\pm 0.5\%$. The F content was measured using fluorine-ion-electrode method; the relative error is $\pm 5\%$. The oxygen content was calculated referring to other components. The ranges of the relative errors are the standard values of the corresponding methods or the corresponding instruments.

3. Results and discussion

The elemental batch composition and analyzed composition are shown in Table 1. The results show that there exists a large difference between batch composition and analyzed composition. Because the loss of P, Pb and Br, the analyzed contents of the other elements are higher than the batch contents.

To make a better understanding of the composition change during the melting process, we examine the relation between the loss percentage of each element and the total element loss portion of a sample. Supposing the total quantity of batch is 100 units (in number of atoms), the analyzed percentage, the batch quantity and the loss quantity of an element have the following relationship:

$$\frac{X_i - \Delta X_i}{100 - \sum \Delta X_i} \times 100 = X'_i, \quad (1)$$

where X_i is the batch amount of an element i , X'_i the analyzed percentage of the element, ΔX_i the loss amount of the element, and $\sum \Delta X_i$ the total loss amount of the sample.

If $\sum \Delta X_i$ is replaced by X_0 , then Eq. (1) will change to:

$$\frac{\Delta X_i}{X_i} \times 100 = 100 - \frac{X'_i}{X_i} (100 - X_0) \quad (2)$$

Table 1
Calculated elemental batch composition and chemical analysis of glasses^a

Glass	Composition/atom (%)					
	Pb	P	F	Cl	Br	O
1 a	16.67	14.29		23.81	9.52	35.71
b	16.1	8.29		19.80	8.12	47.69
2 a	16.67	14.28		19.06	14.29	35.70
b	15.41	7.99		21.6	10.04	44.96
3 a	17.48	13.94		18.59	13.93	36.06
b	16.02	6.23		18.95	9.53	49.27
4 a	17.27	13.77	2.41	18.35	13.77	34.43
b	16.09	6.3	3.24	19.18	8.6	46.59
5 a	17.37	13.84	1.21	18.48	13.86	35.24
b	15.66	6.61	2.98	18.72	10.99	45.04
6 a	14.58	16.41	2.27	8.2	16.40	42.14
b	14.5	7.29	3.29	12.42	10.87	51.63

^a a: Batch composition; b: analyzed composition.

or

$$\Delta X_i = X_i - X'_i \left(\frac{100 - X_0}{100} \right). \quad (3)$$

If we can measure X_0 exactly, we will be able to calculate the values of $((\Delta X_i/X_i) \times 100\%)$ and ΔX_i via Eqs. (2) and (3). However, during the melting process, the incomplete decomposition of $\text{NH}_4\text{H}_2\text{PO}_4$ or the reaction between P_2O_5 and H_2O

in the atmosphere increases the amount of oxygen in the glasses. Therefore, it is very difficult to measure X_0 exactly.

To analyze the relative loss of each element, we plot the relations between $((\Delta X_i/X_i) \times 100\%)$ and X_0 of some samples in Fig. 1. Since there is no cross between the lines in Fig. 1, a line in a higher position indicates that the loss of the element is higher regardless of the value of X_0 . Fig. 1 shows

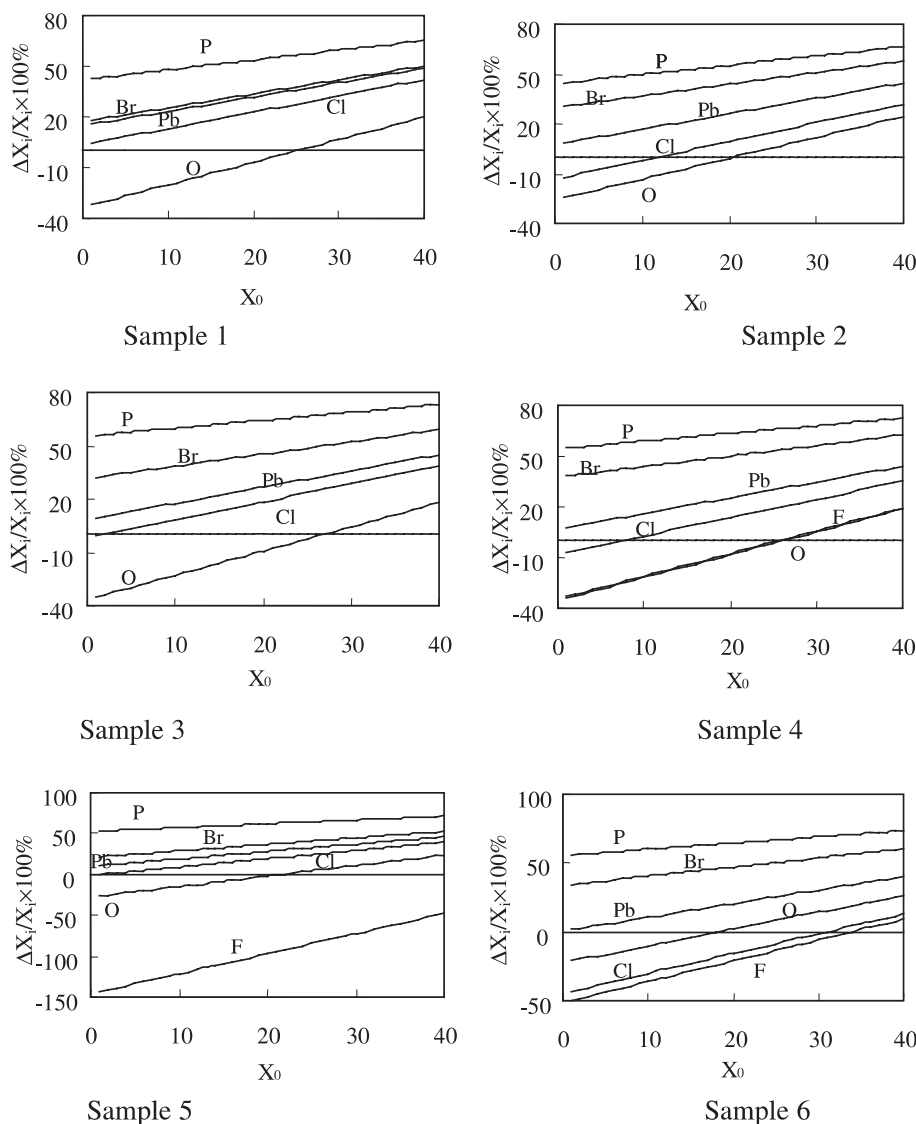


Fig. 1. The relations between $(\Delta X_i/X_i) \times 100\%$ and X_0 of some samples (X_0 , total loss amount of a sample; $(\Delta X_i/X_i) \times 100\%$, loss percentage of i element).

that the curves of P for all the samples are in highest position among those of all the elements, indicating the loss of P is the highest. The loss of the other elements from high to low is that of Br, Pb, Cl and F. The change of oxygen content in the melting process is influenced by two factors. The incomplete decomposition of $\text{NH}_4\text{H}_2\text{PO}_4$ or the reaction between P_2O_5 and H_2O in the atmosphere increases the content of O, while the volatilization of P_2O_5 decreases the content of O.

To further understand whether all the Pb, Cl, and Br are lost in the form of PbCl_2 and PbBr_2 , we examined the ratio of the total loss amount of Cl and Br to that of Pb. According to Eq. (3), the total loss amount of Cl (ΔX_{Cl}) and Br (ΔX_{Br}), the loss amount of Pb (ΔX_{Pb}), and the total loss amount of the sample (X_0) have the following relationship:

$$\frac{\Delta X_{\text{Cl}} + \Delta X_{\text{Br}}}{\Delta X_{\text{Pb}}} = \frac{(X_{\text{Cl}} + X_{\text{Br}}) - (X'_{\text{Cl}} + X'_{\text{Br}})((100 - X_0)/100)}{X_{\text{Pb}} - X'_{\text{Pb}}((100 - X_0)/100)}. \quad (4)$$

Fig. 2 gives the relations between the ratio of $(\Delta X_{\text{Cl}} + \Delta X_{\text{Br}})/\Delta X_{\text{Pb}}$ and the total loss amount X_0 of the samples. It can be seen that, when the total loss amount of the sample X_0 is above 20, the ratio of the total loss amount of Cl and Br to Pb of all

the samples is near 2 regardless the specific value of X_0 . This indicates that the loss of Pb, Br and Cl is mainly in the form of PbBr_2 and PbCl_2 . The ratio for sample 1 is clearly above 2. This may be caused by experimental error.

The transverse axis, X_0 in Fig. 2 begins from 20 since we believe the total loss of a sample X_0 cannot be below 20. We can see from Fig. 1 that, when X_0 is less than 25.6, the losses of F of sample 4, 5, and 6 are all negative. Since the amount of F cannot be increased in the melting process, the total loss amount X_0 of the samples cannot be less than 25.6. Because the compositions and melting processes of the samples without F are similar to those of the samples containing F (the amount of F is small in the samples), we believe that the total loss amount X_0 of samples 1, 2, and 3 are roughly the same as that of samples 4, 5, and 6. Thus, the total loss of a sample X_0 is believed to be over 20.

During the melting process, there may be some reactions between lead halides and P_2O_5 to form phosphorus halides and/or phosphorus oxyhalides, such as PCl_5 , PBr_5 , PBr_3 , POBr_3 , PF_3 etc. If these reactions happen, the total loss amount of Br and Cl will be clearly larger than twice the loss amount of Pb. From the experimental data discussed above, we can conclude that these reactions

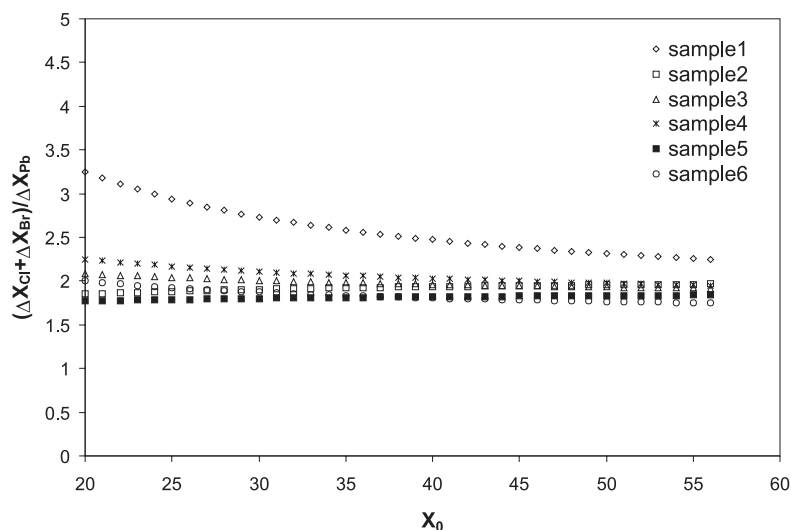


Fig. 2. The relations between the ratio of $(\Delta X_{\text{Cl}} + \Delta X_{\text{Br}})/\Delta X_{\text{Pb}}$ and total loss amount X_0 of the samples.

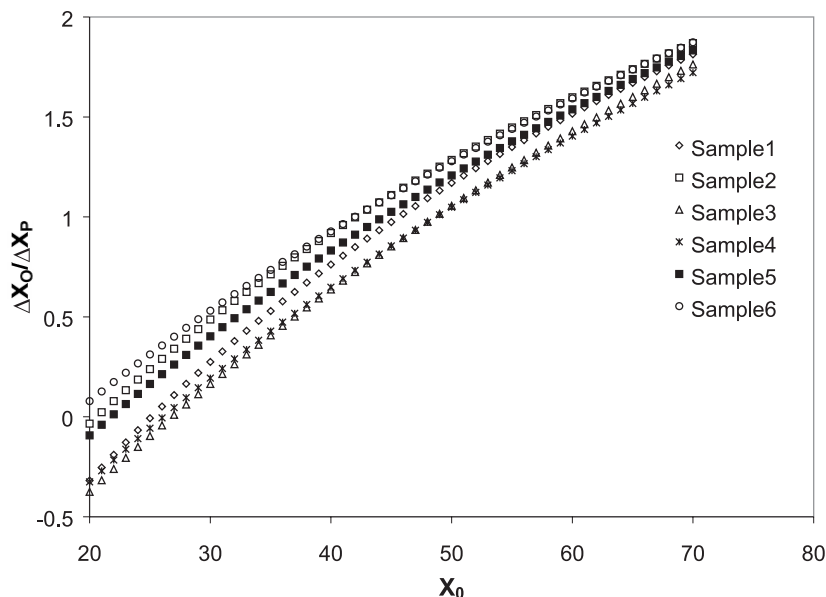


Fig. 3. The relations between the ratio of $\Delta X_O/\Delta X_P$ and total loss amount X_0 of all the samples.

did not take place or did not reach a detectable degree.

According to Eq. (3), we can also get the relationship of the loss amount of O (ΔX_O), the loss amount of P (ΔX_P), and the total loss amount of the sample (X_0)

$$\frac{\Delta X_O}{\Delta X_P} = \frac{X_O - X'_O((100 - X_0)/100)}{X_P - X'_P((100 - X_0)/100)}.$$

Fig. 3 gives the relations between the ratio of $\Delta X_O/\Delta X_P$ and the total loss amount X_0 of the samples. It can be seen that, when the total loss amount is above 20, the ratio of the loss amount of O to P is well below 2.5 and it increases with the total loss amount of the sample. Even if the total loss amount is increased to 70, the ratio of the loss amount of O to P is still below 2. Since P cannot be lost in detectable amount in the forms of phosphorus halides and oxyhalides as discussed above, the loss of P is mainly in the form of P_2O_5 . The fact that the ratio of loss amount of O to P is less than 2.5 indicates that there must be an increase in the

amount of oxygen in the melting process. The mechanisms to cause the increase of oxygen are discussed above.

4. Conclusions

There is a remarkable difference between the batch composition and the analyzed composition of glasses in the $PbBr_2$ – $PbCl_2$ – PbF_2 – PbO – P_2O_5 system. Large amounts of P, Br, Pb, and Cl were lost mainly in the form of $PbBr_2$, $PbCl_2$ and P_2O_5 . The oxygen content is influenced by two factors: The incomplete decomposition of $NH_4H_2PO_4$ or the reaction between P_2O_5 and H_2O in the atmosphere which increases the oxygen content and the volatilization of P_2O_5 which decreases the oxygen content.

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